



How Electronic Architecture In Vehicles Is Revolutionising The Automotive Industry

Over the past few years, the automotive industry has seen a significant increase in the use of electronic systems. These systems have become an essential component of an automobile today. With a focus on autonomous, electric, and connected technologies, vehicles transform into electronic devices on the move. This transformation requires technologies that enable more functionality and power in less space, that is, technologically advanced and compact electronic components



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Every modern car has electronic features incorporated. If we take an example of the recently mandated Automated Braking System (ABS), it features wheel-speed sensors and electronic control units (ECUs). Similarly, assisted driving functions feature LiDAR, radar, camera systems, and much more. As we move towards electric, connected, and autonomous mobility, these components are bound to increase in vehicles.

To put this in perspective, the global market for automotive electronics is estimated to reach \$382.16 billion by 2026. The market is likely to grow at a CAGR of 7.3 per cent from 2019 to 2026, according to a report by Allied Market Research. In 2019, the global automotive electronics market accounted for \$228.34 billion. The earlier, luxury-cars-only electronics has now penetrated all automotive segments.

The automotive industry is working to incorporate all the future-ready features to keep up with the evolving automotive landscape and consumer requirements. This will

eventually pave the way for advanced driving functions, smart technologies, vehicle convenience, comfort, and safety.

Below are a few automotive electronic trends that have transformed the entire vehicle architecture over the past few years.

Electronic control units

While moving towards a more advanced future, the automobile industry will need to incorporate sophisticated features in the vehicles, which require more computation and electronics with high-end technology. These electronic systems are overseen and powered by ECUs. Certain luxury cars have more than 150 ECUs installed to look after various functions, improving the car's efficiency in all respects.

An ECU is a device that oversees, regulates, and alters the operation of a car's electronic system. An ECU will typically control anti-lock braking systems, electronic fuel injection setup, and other electronic features. Some systems have their own ECUs, whereas